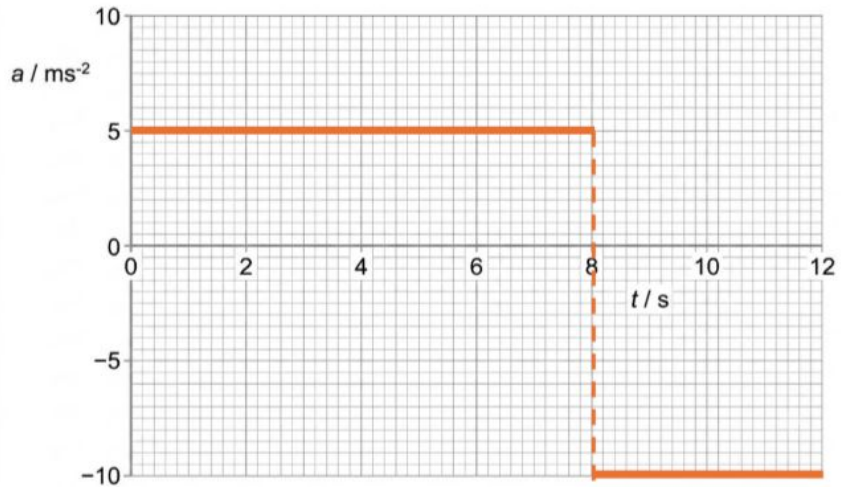
1	(a)	Maximum momentum = 7.2×10^4 N s Maximum KE = $\frac{p^2}{2m} = \frac{(7.2\times10^4)^2}{2\times1800} = 1.4\times10^6$ J	C1 A1
	(b)	v–t graph follows the same shape with maximum speed of 40 m s^{-1} Area under v–t graph = $0.5 \times 40 \times 12 = 240$ m	C1 A1
	(c)	 Gradient of p–t graph = force, Acceleration = force/mass Horizontal line from (0,5.0) to (8.0, 5.0) and horizontal line from (8.0,–10.0) to (12.0,–10.0)	 B1 B1